

The long shadow of COVID-19 on sick leave and primary healthcare consultations in Norway

A narrative review

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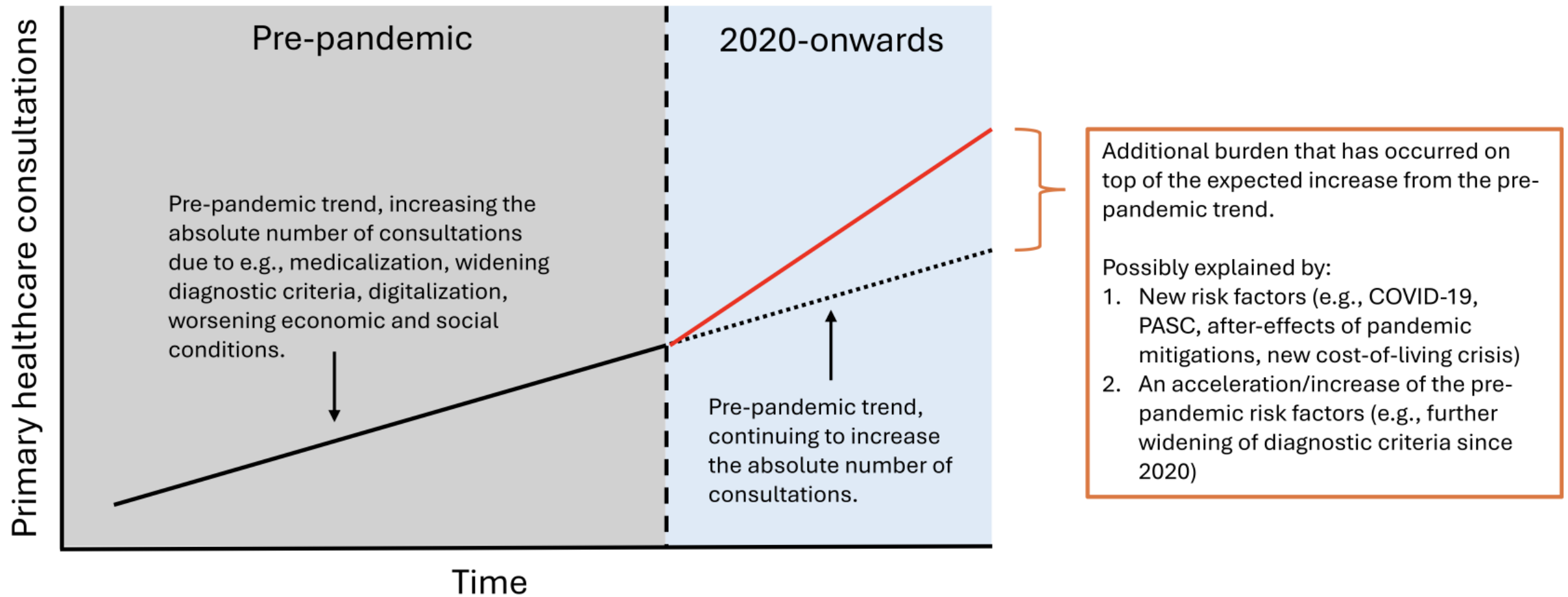
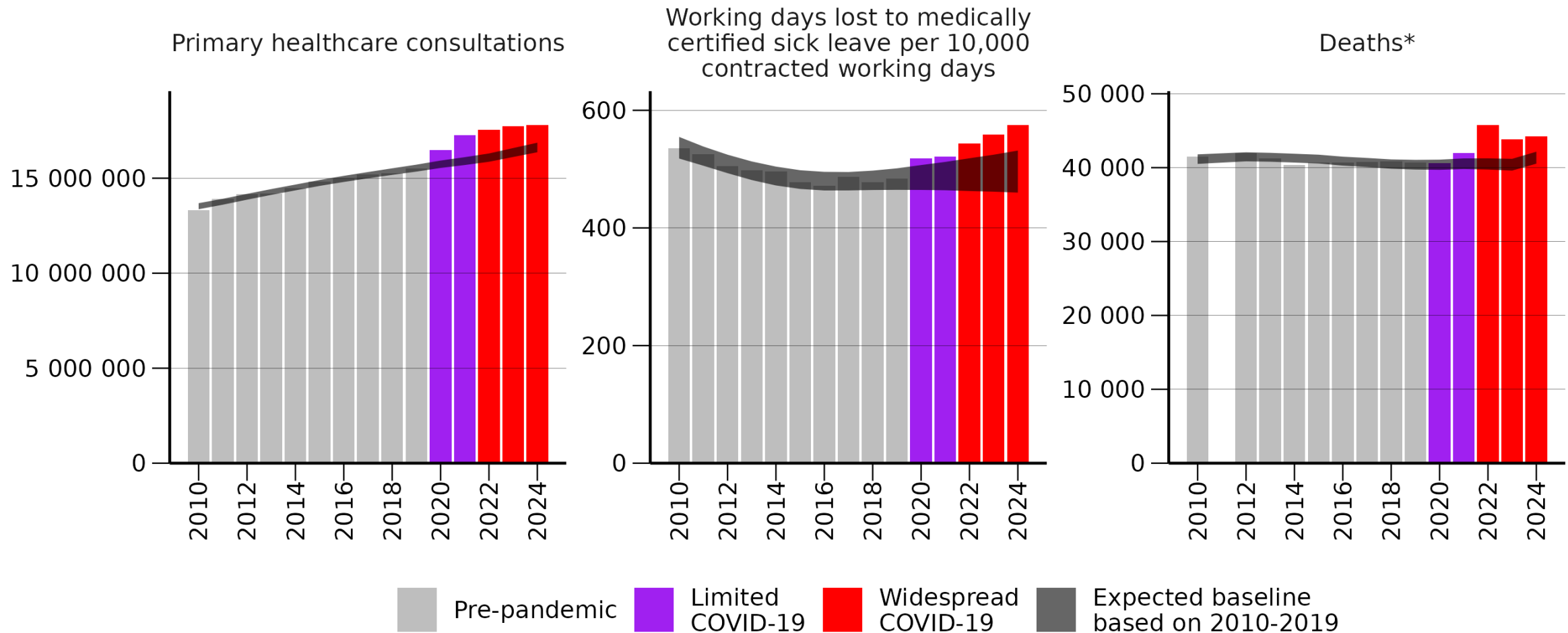


Fig. 5 Conceptual framework for distinguishing pre-pandemic trends from post-2020 increases in primary healthcare consultations

“If COVID is bad, then why does everything seem like 2019?”

- With frequent reinfections (2-4 million/year in Norway?), logically this *should* have an impact on the societal level:
 - 4 million infections per year * 5% risk of long covid = 200.000 per year
- We would primarily expect an impact in the areas of:
 - Fatigue
 - Memory/concentration
 - Immune system

Norway's health indicators are consistently worse than expected, when compared to pre-pandemic trends



Narrative review source: White, R.A. and Nyborg, G.A, The long shadow of COVID-19 on Norwegian sick leave and primary healthcare consultations, 2025, (submitted)

A) White RA, Salamanca BV, Angelsen A, Zakiudin DP, Andries A, Nyborg GA. Excess primary healthcare consultations in Norway in 2024 compared to pre-COVID-19-pandemic baseline trends 2025 (submitted)

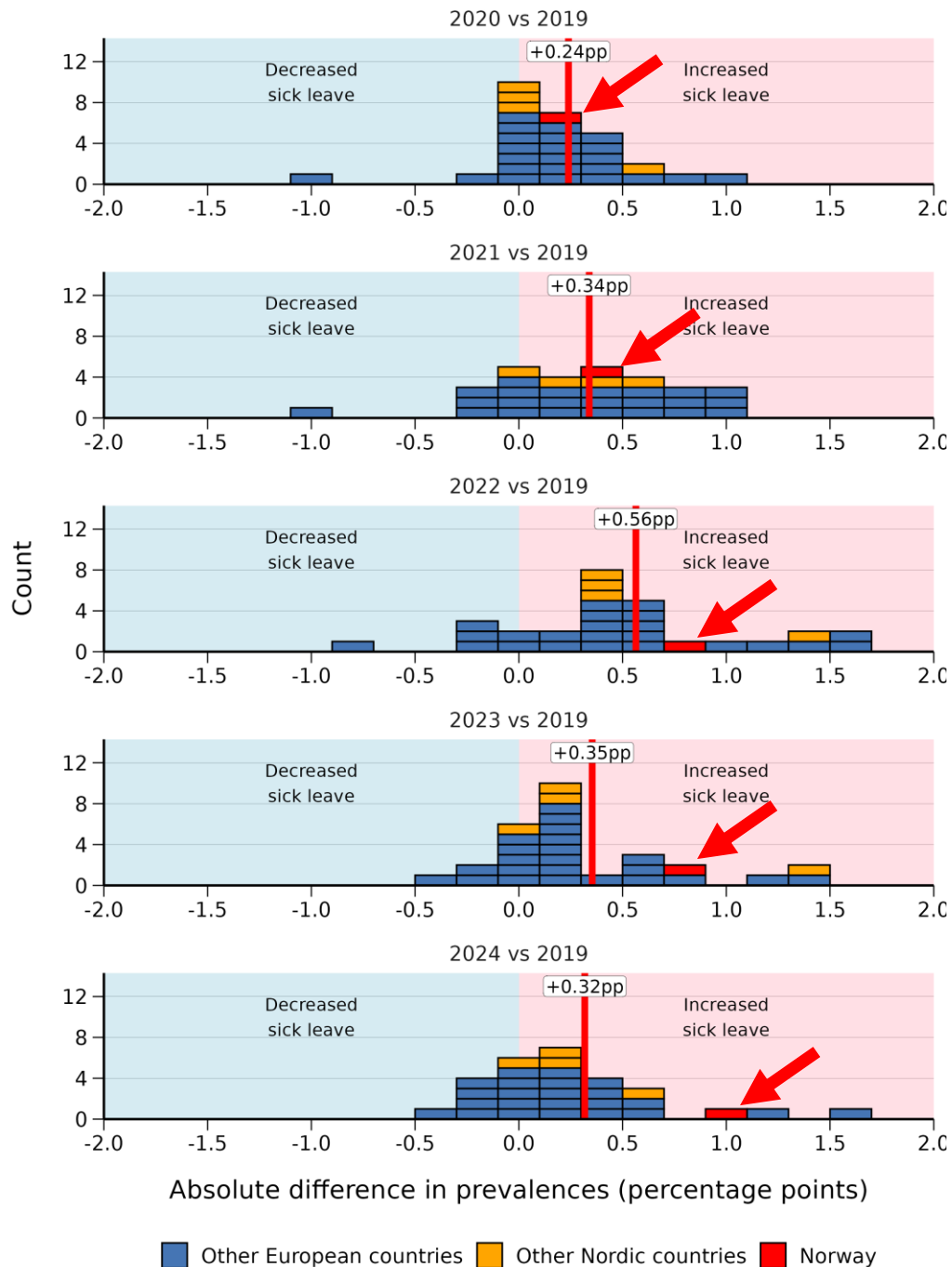
B) White RA, Zhang C, Valcarcel Salamanca B, Angelsen A, Zakiudin DP, Andries A, et al. Aberrations in medically certified sick leave and primary healthcare consultations in Norway in 2023 compared to pre-COVID-19-pandemic trends. Arch Public Health 2024;82:187. <https://doi.org/10.1186/s13690-024-01411-4>.

C) White RA, Nygaard AB, Søråas A, Nyborg GA. Excess all-cause mortality in Norway in 2024. Scand J Public Health 2025;0:14034948251371830.

Sick leave across Europe

Comparing against 2019 baseline:

- European sick leave peaked in 2022.
- However, European sick leave is still elevated in 2024!
- Norwegian sick leave worsening every year.

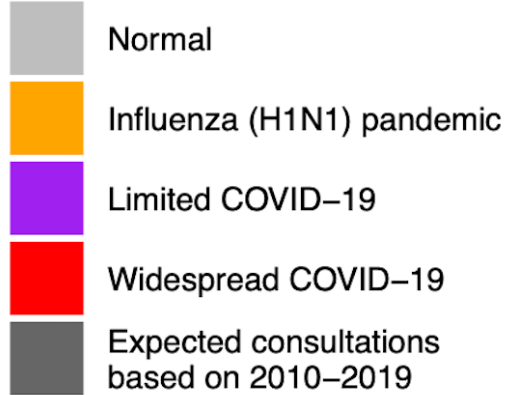


Note that there were methodological changes between 2020 and 2021.

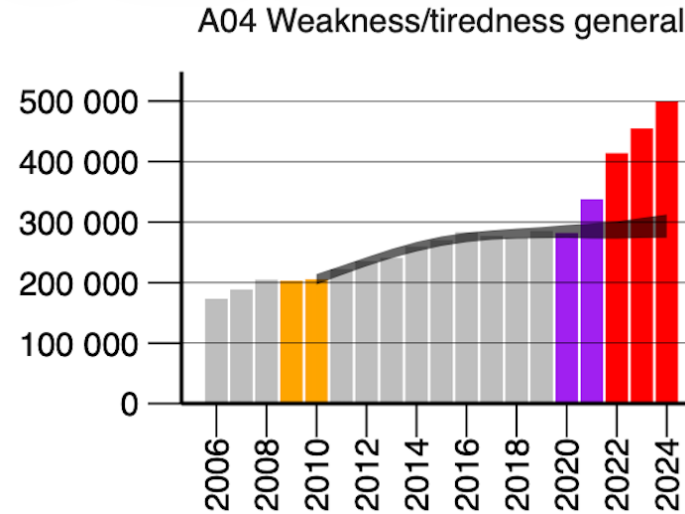
Narrative review source: White, R.A. and Nyborg, G.A, The long shadow of COVID-19 on Norwegian sick leave and primary healthcare consultations, 2025, (submitted)

A) Eurostat. EU - Labour Force Survey microdata 1983-2023, release 2024 2024.
<https://doi.org/10.2907/LFS1983-2023>.

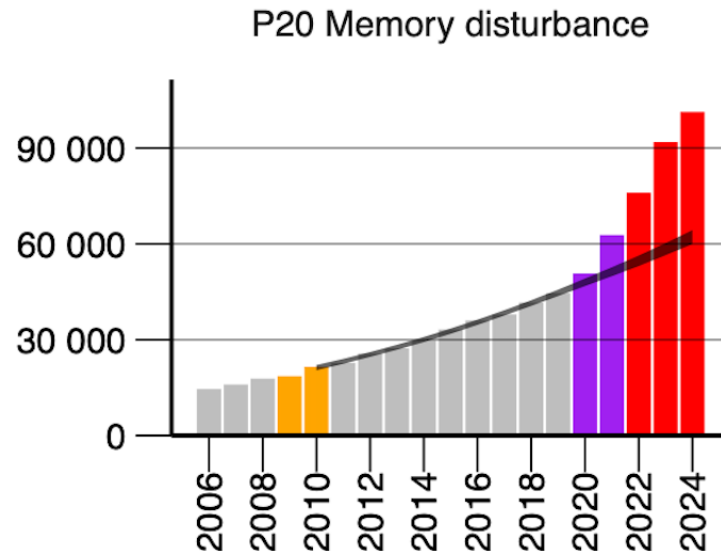
Primary healthcare consultations



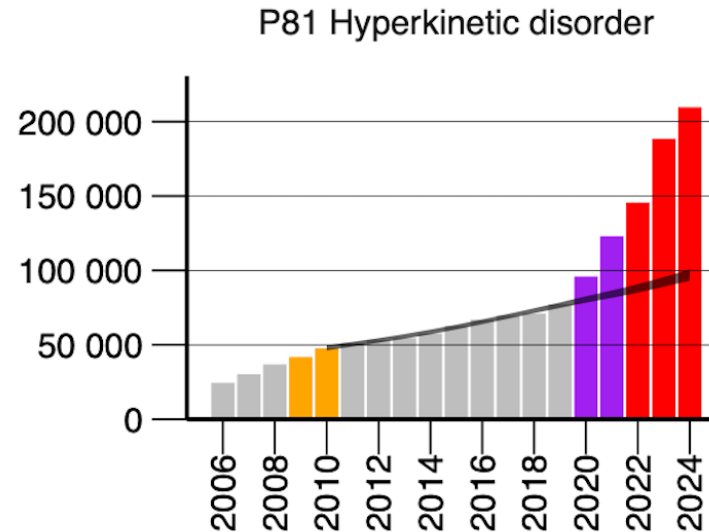
Fatigue



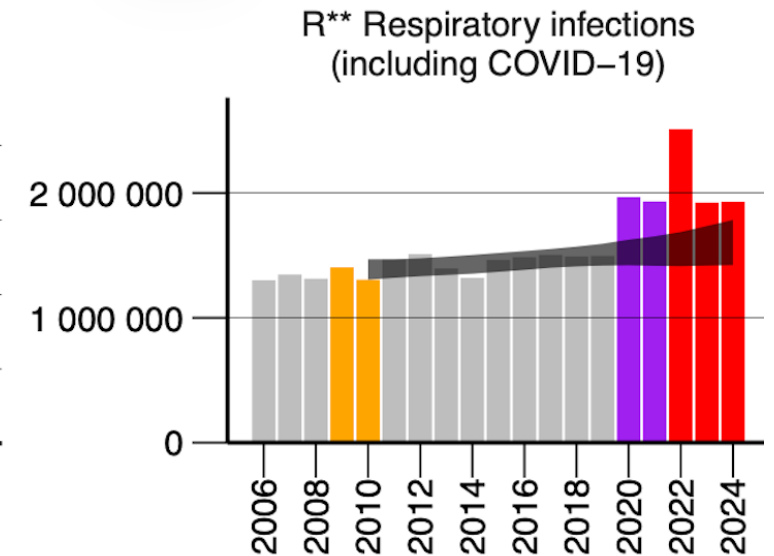
Memory



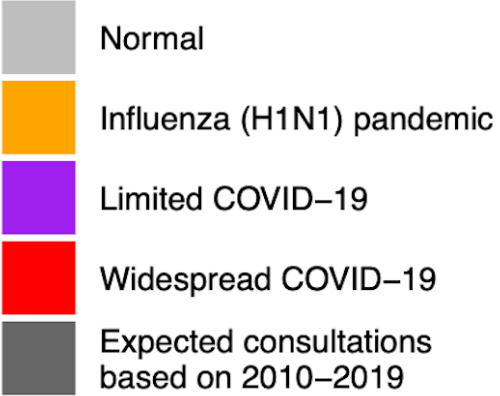
Concentration



Immune system



Primary healthcare consultations

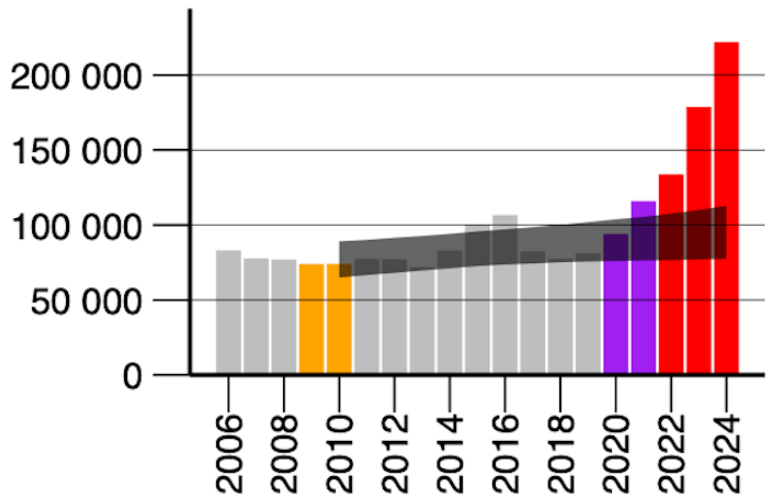


Depressive psychological issues**

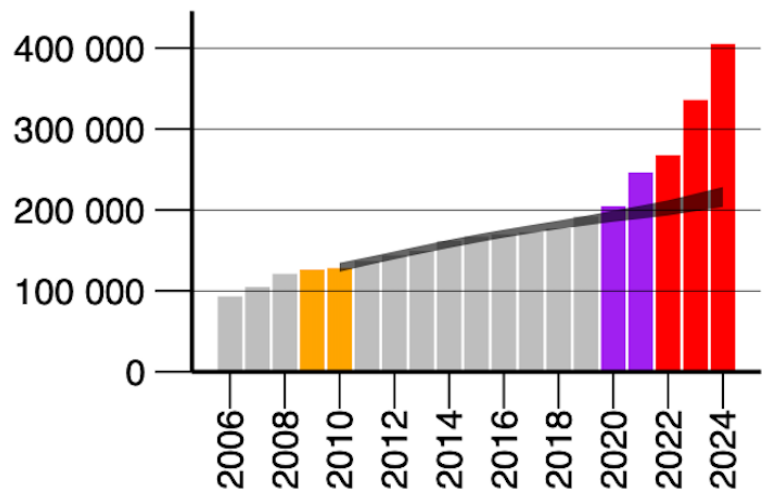
Non-depressive psychological issues**

Non-viral fatigue/burnout

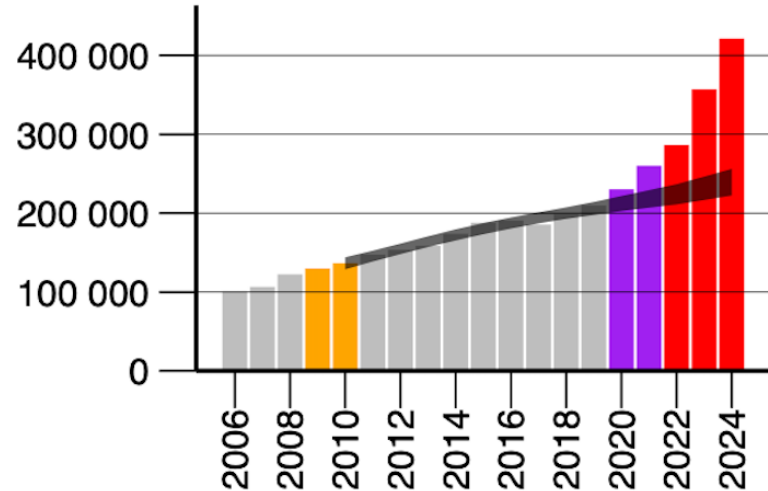
P03 Feeling depressed



P29 Psychological symptom/complaint other



P02 Acute stress reaction



White RA, Salamanca BV, Angelsen A, Zakiudin DP, Andries A, Nyborg GA. Excess primary healthcare consultations in Norway in 2024 compared to pre-COVID-19-pandemic baseline trends 2025 (submitted)

Identification of studies via databases and registers

Identification of studies via other methods

Identification

Records identified from:
PubMed (n = 18)

Records removed *before screening*:
Duplicate records removed (n = 2)
Records marked as ineligible by automation tools (n = 0)
Records removed for other reasons (n = 0)

Records identified from:
NAV "Sick leave analysis website" (n = 6)
NAV "Annual reports" (n=1)
Citation searching (n = 13)
Other (n=2)

Screening

Records screened
(n = 16)

Records excluded manually
(n = 13)

Reports sought for retrieval
(n = 3)

Reports not retrieved
(n = 0)

Reports assessed for eligibility
(n = 3)

Reports excluded:
No non-covid controls (n=1)

Reports sought for retrieval
(n = 22)

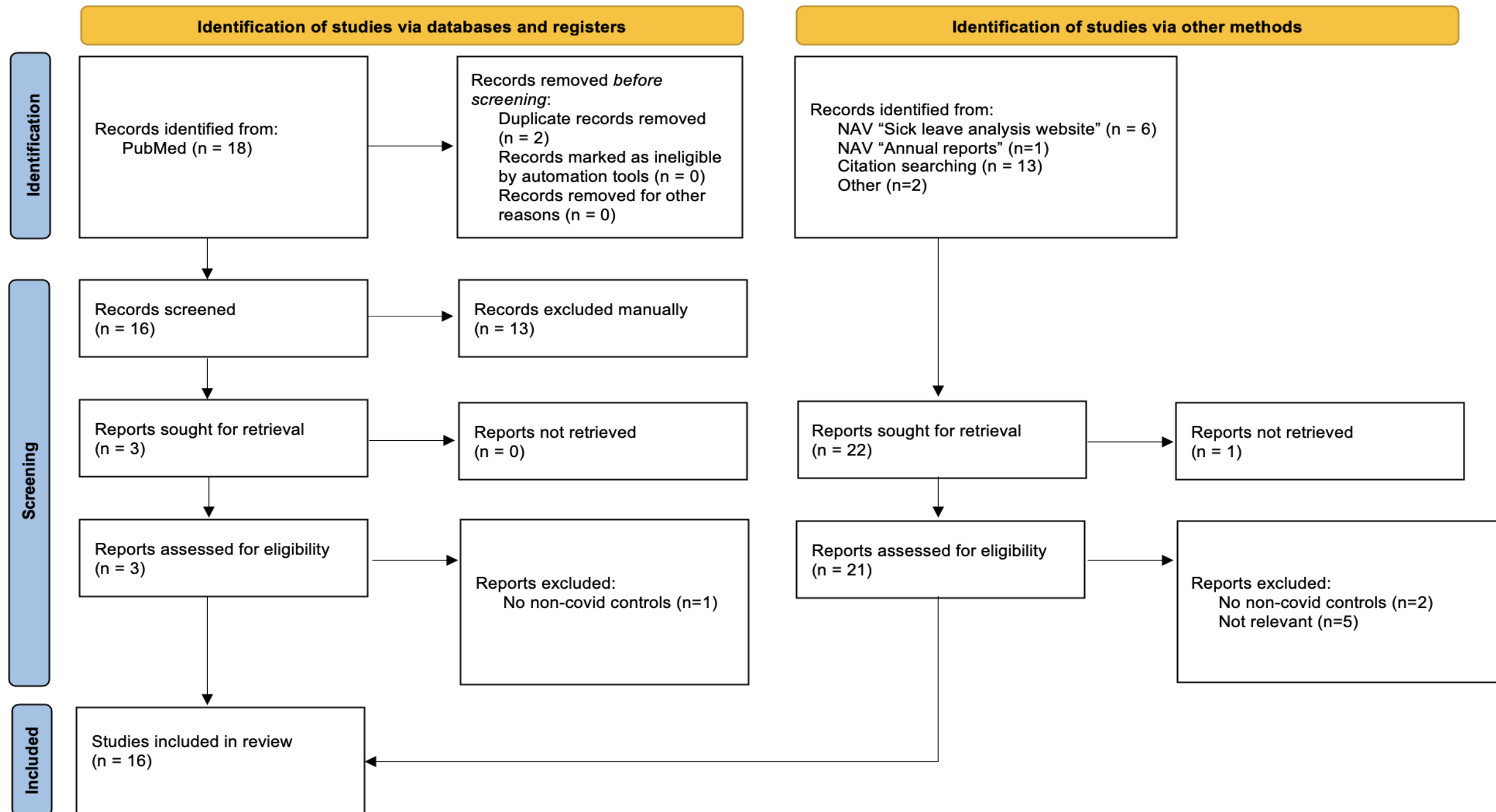
Reports not retrieved
(n = 1)

Reports assessed for eligibility
(n = 21)

Reports excluded:
No non-covid controls (n=2)
Not relevant (n=5)

Included

Studies included in review
(n = 16)



Primary healthcare consultations				
Respiratory	Fatigue (A04)	Memory (P20) and concentration (P81)	Psychological issues other than depression (P02, P29)	Feeling depressed (P03)
Association between COVID-19 and increased risk in Norwegian registry studies at the person-level				
	Yes [45,46]	P20: Yes [45,46] P81: Untested	Untested	No [45]
Number of consultations in 2024 expected from pre-pandemic trends [36]				
1,602,392	293,353	P20: 62,116 P81: 96,846	P02: 239,229 P29: 216,262	95,188
Number of consultations recorded in 2024 [36]				
1,928,118	498,734	P20: 101,293 P81: 209,609	P02: 421,308 P29: 405,240	221,971
Percent higher in 2024 than expected from pre-pandemic trends [36]				
20%	70%	P20: 63% P81: 116%	P02: 76% P29: 87%	133%
Percent of the total excess in 2024 explained by this ICPC-2 code [36]				
27%	17%	P20: 3% P81: 10%	P02: 15% P29: 16%	11%
Temporal correlation with COVID-19 community spread (2020-2024) [36]				
Yes	Yes	P20: Yes P81: Yes	P02: No P29: No	No

Medically certified sick leave				
Respiratory	Fatigue (A04)	Memory (P20) and concentration (P81)	Psychological issues other than depression (P02, P29)	Feeling depressed (P03)
Association between COVID-19 and increased risk in Norwegian registry studies at the person-level				
Yes [49]	Yes [49]	Untested	P02: Yes [49] P29: Yes [49]	P03: Untested P73: No [49]
Working person-days lost to sick leave in 2023 expected from pre-pandemic trends [37]				
1,794,000	824,000	P20: N/A P81: 65,000	P02: 1,377,000 P29: 980,000	129,000
Working person-days lost to sick leave in 2023 [37]				
2,646,000	1,484,000	P20: N/A P81: 123,000	P02: 1,896,000 P29: 1,707,000	406,000
Percent higher in 2023 than expected from pre-pandemic trends [37]				
47%	80%	P20: N/A P81: 91%	P02: 38% P29: 74%	214%
Percent of the total excess in 2023 explained by this ICPC-2 code [37]				
14%	11%	P20: N/A P81: <1%	P02: 8% P29: 12%	5%
Temporal correlation with COVID-19 community spread (2020-2023) [37]				
Yes	Yes	P20: Untested P81: Yes	P02: Yes P29: Yes	Yes

Table 3. Synthesis of evidence pertaining to the ICPC-2 codes responsible for the largest contributors

While other factors undoubtedly also play a role, the convergence of individual-level findings with population-level surveillance data is compatible with the hypothesis of sustained health and economic impacts from widespread SARS-CoV-2 reinfections as of 2024.

If this hypothesis is correct, then Norway needs to explicitly acknowledge and include PASC and reinfection-associated morbidity in policy assessments.